

stage2_multi_cli_comparison

Comprehensive Comparative Analysis of TOP 10 CLI AI Coding Agents

For HelixCode Integration - Stage 2 Research Report

Date: May 2026
Scope: 10 Leading CLI AI Coding Agents (excluding Claude Code)
Methodology: GitHub API analysis, README/code review, architecture documentation, web research

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1. Executive Summary

This report analyzes the top 10 open-source CLI AI coding agents to identify the best features, architectural patterns, and innovations that should be ported to HelixCode. Each agent was evaluated across 11 dimensions: Repository Structure, Core Features, Unique Innovations, Architecture Patterns, Tool Use, Context Management, Edit Format, UI Components, Performance Features, Integration APIs, and Game Changers.

Key Findings at a Glance

Agent	Stars	Lang	License	Best At	Killer Feature
Gemini CLI	103K	TypeScript	Apache-2.0	Large context, research	1M token context + Plan Mode
OpenAI Codex	80K	Rust	Apache-2.0	Production agent loop	Sandboxed execution + ZDR
Cline	61K	TypeScript	Apache-2.0	IDE integration, control	Plan/Act + Checkpoints
GPT Engineer	55K	Python	MIT	Full project generation	Prompt-based scaffolding

Agent	Stars	Lang	License	Best At	Killer Feature
Aider	44K	Python	Apache-2.0	Git-native pair programming	Architect/Editor dual-model
Kilo Code	19K	TypeScript	MIT	Multi-mode orchestration	5 agent modes + subagents
Plandex	15K	Go	MIT	Large-scale refactoring	2M token sandboxed diffs
OpenCode	12K	Go	MIT	BYOK freedom, plugins	LSP + 75+ providers
Forge	7K	Rust	Apache-2.0	Multi-agent orchestration	6 orchestration patterns
Amazon Q	2K	Rust	Apache-2.0	AWS ecosystem	Fig integration + diagrams

2. Agent-by-Agent Deep Dive

2.1 Aider (Aider-AI/aider) - 44,304 stars

Repository Structure: - Language: Python (100%) - Size: 140MB - Monolithic Python package with `aider/` as main module - `website/docs` in `aider/website/` - CLI entry via `aider/main.py`

Core Features: - Git-native pair programming: every change auto-committed - Support for 100+ programming languages - Voice-to-code input (`/voice` command) - Image input for UI development - IDE watch mode (file changes trigger AI) - Architect mode + Editor model dual-model workflow - 4 edit formats: diff, udiff, whole, diff-fenced - Repo map via Tree-sitter for codebase understanding - Auto test/lint-fix loop - Benchmark-leading accuracy (81-88% on polyglot benchmarks)

Unique Innovations: 1. **Architect/Editor Mode:** Separates reasoning (architect model like o1/o3) from editing (editor model like GPT-4o/Claude). Architect proposes solution, editor converts to actual file edits. Benchmarks show 85% pass rate with o1-preview + o1-mini. 2. **Unified Diff Format:** Created specifically to combat GPT-4 Turbo “lazy coding” (eliding code with “# ... original code here ...”). Makes GPT-4 Turbo 3X less lazy. 3. **Flexible Search/Replace Matching:** 4-layer matching strategy (exact, whitespace-insensitive, indentation-preserving, fuzzy via difflib) for robust edit application. 4. **Repo Map:** Tree-sitter-based AST parsing to build a compressed representation of codebase structure (classes, functions, imports) within token budget (default 1k map tokens).

Architecture Patterns: - Single-agent ReAct loop with tight git integration - Prompt caching optimization through careful prefix ordering - Hierarchical context: repo map → added files → conversation history - Git as session store: every action committed, full history inspectable

Tool Use: - No MCP (Model Context Protocol) - custom tool set - Built-in tools: `read_file`, `write_file`, `search/replace`, `shell command`, `git operations` - LSP integration for language-specific context - Browser automation via Playwright (optional)

Context Management: - Manual /add and /drop for file context control - /tokens command shows approximate token count and cost per file - Repo map auto-generated for all tracked files - /reset drops all context - Context window managed by user with granular control

Edit Format: - **diff** (search/replace blocks): Primary format, most models - **udiff** (unified diff): For GPT-4 Turbo, prevents laziness - **whole** (full file rewrite): Fallback for weak models - **diff-fenced** (path-in-fence): For Gemini models - **editor-diff/editor-whole**: Streamlined versions for architect mode

UI Components: - CLI terminal interface (rich text via Python Rich library) - VS Code extension (companion) - Voice input mode - Streaming output with syntax highlighting

Performance Features: - Prompt caching via prefix preservation - Lazy loading of repo map - Only sends changed files in context - Model-optimized edit format selection

Integration APIs: - Command-line interface with extensive flags - .aider.conf.yml for project configuration - .aiderignore for exclusion patterns - Environment variables for model selection

Game Changers: 1. **Architect/Editor dual-model** - separates reasoning from editing, dramatically improving quality 2. **4-layer fuzzy matching** for search/replace - makes edits robust even with imperfect AI output 3. **Git-native workflow** - every change auto-committed with meaningful messages, making AI work auditable and reversible

Best Feature to Port: The **Architect/Editor dual-model architecture** - this is a proven pattern that boosts accuracy by separating problem-solving from file manipulation. Any model that can reason but struggles with precise edits benefits enormously.

2.2 Cline (cline/cline) - 61,341 stars

Repository Structure: - Language: TypeScript (100%) - Size: 390MB - VS Code extension architecture with sidebar panel - Core in src/core/, webview UI in src/core/webview/ - CLI 2.0 as separate terminal-native build

Core Features: - Human-in-the-loop: every file change and terminal command requires explicit approval - Browser automation via Puppeteer (Computer Use) - Workspace checkpoints (shadow Git repository) - Plan/Act dual-mode system - Deep Planning (/deep-planning) for complex tasks - Focus Chain (automatic todo list tracking) - Memory Bank (structured documentation for cross-session context) - 30+ LLM providers supported - Cost tracking (token count and API spend per task) - .clinerules/ for project-specific governance - MCP Marketplace for tool discovery

Unique Innovations: 1. **Plan/Act Mode Split:** Plan mode is read-only (can explore codebase but not modify). Act mode executes changes. Context carries over between modes. Can use DIFFERENT models for each mode (e.g., Claude Opus for Plan, Grok Fast for Act). 2. **Checkpoints (Shadow Git):** Independent Git repository that snapshots workspace state after each AI operation. Granular rollback without polluting main Git history. Preserves terminal states and browser sessions. 3. **Computer Use (Browser):** Launches real Chrome, navigates, screenshots, interacts with UI. Can verify UI changes rendered correctly, debug runtime errors visually. 4. **Focus Chain:** Automatic todo list that tracks implementation progress against plan with visible checklist. 5. **Memory Bank:** Structured project documentation that persists across sessions (projectBrief.md, techContext.md, activeContext.md, progress.md).

Architecture Patterns: - VS Code extension as primary surface, CLI 2.0 as terminal-native alternative - Agent Client Protocol (ACP) for editor-agnostic communication - State machine: Plan [?] Act with context preservation - Shadow Git for checkpointing

Tool Use: - Full MCP (Model Context Protocol) support - stdio and SSE transport - Custom tools: `execute_command`, `read_file`, `write_file`, `replace_in_file`, `search_files`, `list_files`, `list_code_definition_names`, `browser_visit`, `ask_followup_question`, `attempt_completion` - MCP Marketplace for one-click tool installation

Context Management: - Full file reading in Plan mode for comprehensive understanding - Memory Bank files for cross-session persistence - Timeline feature for file change tracking - Context carried between Plan and Act modes

Edit Format: - `replace_in_file` with SEARCH/REPLACE blocks (XML-like tags) - Diff view in editor for human review - Can edit or revert changes directly in diff view

UI Components: - VS Code sidebar panel (primary) - JetBrains, Cursor, Windsurf, Zed, Neovim (via ACP) - CLI 2.0 terminal TUI (tab between Plan/Act) - Markdown rendering inline in terminal - Slash commands (`/history`, `/settings`, `/deep-planning`)

Performance Features: - Model switching mid-session (Plan vs Act optimization) - Checkpoints enable safe experimentation - “Proceed While Running” for long-running processes - Lazy Teammate Mode and YOLO mode for autonomy tuning

Integration APIs: - VS Code extension API - CLI 2.0 with `-y` headless mode, `--json` structured output - ACP (Agent Client Protocol) compliance - Custom workflows via Markdown files in `~/ .cline/workflows/`

Game Changers: 1. **Plan/Act with different models** - Using cheap/fast models for execution and expensive/reasoning models for planning optimizes cost and quality 2. **Shadow Git Checkpoints** - Granular rollback system that preserves full environment state independently of user's Git 3. **Computer Use / Browser automation** - Real browser verification bridges static code analysis and runtime behavior gap

Best Feature to Port: The **Plan/Act dual-mode with model switching** - this is a workflow innovation that dramatically improves control and cost-efficiency. Every serious agent should have a read-only planning phase.

2.3 OpenAI Codex (openai/codex) - 79,852 stars

Repository Structure: - Language: Rust (95.6%), with TypeScript bindings - Size: 423MB - Monorepo: `codex-cli/` (main CLI), `codex-rs/` (Rust rewrite) - 640+ tagged releases since April 2025 - 4,000+ commits, 400 contributors

Core Features: - Terminal-native agent with TUI - ChatGPT plan integration (Plus/Pro/Team/Edu/Enterprise) - API key authentication alternative - OS-level sandboxed execution - Enterprise proxy support (custom CA certificates) - Hooks system (user prompt hook, pre/post hooks) - Python SDK and app-server for programmatic access - CI-friendly sandbox - Remote test workflows - Cross-platform binaries (macOS ARM/x86, Linux x86/ARM, Windows native)

Unique Innovations: 1. **Production-Grade Agent Loop:** Stateless request architecture for Zero Data Retention (ZDR) compliance. Deliberately avoids `previous_response_id` to maintain statelessness. 2. **Prompt Caching Optimization:** Careful prefix preservation to achieve linear rather than quadratic performance. Static content (instructions, tools) at beginning, variable

content at end. 3. **Automatic Context Compaction:** /responses/compact endpoint with specialized compaction items containing opaque encrypted content preserving model's latent understanding. 4. **Tiered Sandbox:** Suggest (read-only auto) → Auto-Edit (auto-apply patches) → Full Auto (complete autonomy within sandbox). OS-level enforcement: macOS Seatbelt, Linux Docker+iptables, Windows native sandbox. 5. **Hierarchical Instruction System:** AGENTS.md/AGENTS.override.md cascading from \$CODEX_HOME to CWD with 32 KiB limit. Skills instructions layered on top.

Architecture Patterns: - Rust-based harness (95.6% of codebase) for performance - Agent loop: inference → tool call → execution → observation → repeat - Stateless requests for ZDR compliance - SSE streaming with internal event republishing - Multi-surface: CLI, VS Code extension, Cursor, Windsurf

Tool Use: - Uses Responses API (not Chat Completions) - Codex-provided tools (sandboxed) + API-provided tools + MCP servers - Shell-centric design: one primary tool (shell executor) for reading, searching, editing via `apply_patch` - `apply_patch` for unified diffs (not whole-file rewrites)

Context Management: - Hierarchical prompt construction: system → tools → instructions → input - Automatic compaction when `auto_compact_limit` exceeded - Developer-role messages for sandbox config changes (preserves cache prefix) - AGENTS.md for project-specific instructions

Edit Format: - `apply_patch` (unified diff) - primary format - Minimal surgical diffs rather than whole-file rewrites - Colorized diffs for user review - Explicit approval/deny/edit options

UI Components: - Terminal TUI (interactive) - `--output-format json` for structured output - `--output-format stream-json` for real-time event streaming - Non-interactive mode for scripts (`-p` for simple text)

Performance Features: - Prompt caching through prefix preservation (linear vs quadratic) - Careful tool ordering to avoid cache misses - Context window management via automatic compaction - Rust implementation for speed

Integration APIs: - Python SDK for programmatic embedding - Hooks system for custom integrations - `--oss` flag for local models via Ollama/LM Studio - Enterprise proxy via `SSL_CERT_FILE` environment variables

Game Changers: 1. **Stateless ZDR-compliant architecture** - Proves you can build production agents without storing conversation history server-side 2. **Automatic context compaction with encrypted latent understanding** - Revolutionary approach to context window management that preserves model comprehension 3. **OS-native sandboxing** (Seatbelt/Docker/Windows) - Sets the security standard for CLI agents; every agent should sandbox at OS level

Best Feature to Port: The **OS-native sandboxed execution** - this is the gold standard for agent security. Sandboxing at the OS level (not just application-level permissions) is essential for trustworthy autonomous agents.

2.4 Plandex (plandex-ai/plandex) - 15,326 stars

Repository Structure: - Language: Go (100%) - Size: 57MB - Client-server architecture - CLI client in `app/cli/`, server in `app/server/` - Single binary for CLI, requires server component

Core Features: - 2M token effective context window per session - 20M+ token directory indexing via Tree-sitter - Cumulative diff review sandbox (changes accumulate, apply atomically) - Full auto mode for autonomous execution - REPL mode with fuzzy auto-complete - Client-server architecture (can self-host server) - Model packs with version-controlled settings - Git integration with auto commit messages - Multi-model via OpenRouter API

Unique Innovations: 1. **Cumulative Diff Review Sandbox:** ALL AI-generated changes accumulate in a sandbox separate from project files. Review and test ALL changes before applying ANY. This is critical for multi-session work spanning many files. 2. **2M Token Context Window:** Direct support for 2M tokens (~100k per file), with tree-sitter project maps for 20M+ token directories. One of the largest context windows in CLI agents. 3. **Model Packs with Version Control:** Model settings and entire model packs are version-controlled within Plandex. Can experiment with configurations and revert to stable settings. 4. **Client-Server Architecture:** Server handles model inference, client handles local file operations. Enables self-hosting and cloud usage.

Architecture Patterns: - Client-server: CLI client  HTTP server - Tree-sitter indexing pipeline: parse → extract → store → serve - Sandbox pattern: all edits staged before application - REPL with command history and fuzzy matching

Tool Use: - Custom tool system (not MCP) - Tools: file read/write, shell execution, git operations, codebase search - Tree-sitter-based code understanding

Context Management: - 2M token direct context (massive) - Tree-sitter project maps for 20M+ token directories - Manual file loading per task - Context persists across server sessions

Edit Format: - Cumulative diffs (sandboxed) - Batch application (apply all or none) - Review interface before application

UI Components: - Terminal REPL interface - Fuzzy auto-complete for commands - Diff review interface - No IDE integration (terminal-only by design)

Performance Features: - Tree-sitter incremental indexing - Server-side context management - Efficient binary protocol between client/server

Integration APIs: - HTTP API for server - OpenRouter integration for model access - Self-hosting via Docker

Game Changers: 1. **Cumulative diff sandbox** - The ability to accumulate, review, and apply ALL changes atomically is unique and solves the “AI rewrote half my project” problem 2. **2M token context + 20M token indexing** - Massive context handling for truly large-scale refactoring 3. **Version-controlled model packs** - Treating model configuration as versioned artifacts enables safe experimentation

Best Feature to Port: The **Cumulative Diff Review Sandbox** - this safety mechanism should be standard in every agent. The ability to batch-review AI changes before application prevents catastrophic mistakes.

2.5 Forge (antinomyhq/forge) - 7,177 stars

Repository Structure: - Language: Rust (100%) - Size: 18MB - Modular Rust crates structure - CLI in `crates/forge_main/`, engine in `crates/forge_engine/` - YAML-based configuration (`forge.yaml`)

Core Features: - Multi-agent AI coding orchestrator - 6 orchestration patterns (sequential, parallel, leader-worker, dynamic routing, etc.) - Multi-model coordination (Claude, Gemini, Copilot, Antigravity) - Agent hierarchy: Mayor (coordinator), Polecat (worker), Refinery (reviewer), Triage (resolver) - Quality scoring (100pt scale: structure/code/tests/docs) - A/B testing for prompts and agent combos - HTML dashboard with Chart.js - Resume/save-restore pipeline state - Benchmark runner with standard objectives

Unique Innovations: 1. **6 Orchestration Patterns:** Sequential Pipeline, Parallel Fan-Out, Leader-Worker, Dynamic Routing, Explorer-Critic, Iterative Kanban. This is the most sophisticated multi-agent orchestration in CLI tools. 2. **Agent Hierarchy with Specialized Roles:** Mayor (persistent coordinator), Polecat (clones repos, writes code, commits, pushes), Refinery (reviews branches, runs quality gates), Triage (resolves ambiguities). 3. **Quality Scoring System:** 100-point scale evaluating structure, code quality, tests, and documentation. Enables automated quality gates. 4. **A/B Testing Framework:** Built-in framework for comparing prompts and agent combinations with statistical significance.

Architecture Patterns: - Multi-crate Rust workspace - Async engine with parallel dispatch - Pipeline phases: PLAN → CODE → VERIFY → REVIEW → FIX - Reconciler loop (every 5 seconds) driving state transitions - Pydantic-based configuration validation

Tool Use: - MCP support for external tools - Built-in tools: bash, file operations, git, search - Plugin registry with hook-based extension

Context Management: - Smart file chunking and context windowing - Session + persistent cross-run memory - Workspace-aware context gathering - Context compacting for long sessions

Edit Format: - Patch-based edits - Diff application - Whole file writes for scaffolding

UI Components: - Terminal UI with Rich library panels - HTML dashboard for monitoring - TUI for interactive sessions

Performance Features: - Parallel agent execution - Async/await throughout - Smart context windowing - Resume/save-restore for long tasks

Integration APIs: - YAML configuration (`forge.yaml`) - CLI with multiple subcommands - Plugin system with hooks

Game Changers: 1. **6 orchestration patterns** - The most complete multi-agent orchestration toolkit available; enables complex workflows impossible with single agents 2. **Quality scoring with automated gates** - 100pt scale with automated pass/fail enables CI/CD integration 3. **A/B testing for agent configurations** - Data-driven approach to finding optimal prompt+model combinations

Best Feature to Port: The **6 orchestration patterns** - Sequential, Parallel, Leader-Worker, Dynamic Routing, Explorer-Critic, and Kanban. This is the future of agentic development and HelixCode should support multiple coordination patterns natively.

2.6 Kilo Code (Kilo-Org/kilocode) - 18,881 stars

Repository Structure: - Language: TypeScript (primary) - Size: 407MB - Central CLI engine (`packages/opencode/`) - VS Code extension (`packages/kilo-vscode/`) - Kilo Gateway (`packages/kilo-gateway/`) - KiloClaw cloud platform

Core Features: - 5 specialized agent modes: Ask, Architect, Code, Debug, Orchestrator - Subagent delegation (agents can spawn child agents) - 500+ models via provider router - Memory Bank for cross-session context - KiloClaw cloud agents (24/7 autonomous) - GitHub/GitLab bot integration - Auto Triage (issue classification, duplicate detection) - Code Review service (automated PR reviews) - App Builder (full app generation from prompts) - Gas Town multi-agent orchestration platform - JetBrains support (all plans, not just Enterprise)

Unique Innovations: 1. **5 Specialized Modes:** Each mode has its own prompt, tools, and context policy. Architect plans, Code implements, Debug fixes, Ask answers, Orchestrator coordinates. This is like having 5 different agents in one tool. 2. **Subagent Delegation:** Any agent with full tool access (Code, Plan, Debug) can launch subagents automatically via `task` tool. Subagents run in isolated context, return summaries to parent. Enables parallel exploration. 3. **KiloClaw / Gas Town:** Multi-agent orchestration on Cloudflare. Town (workspace) → Rig (repo) → Bead (task) → Convoy (batch). Agents: Mayor (coordinator), Polecat (worker), Refinery (reviewer), Triage (resolver). Reconciler loop every 5 seconds. 4. **Auto Triage:** Classifies GitHub issues (bug/feature/question), detects duplicates via vector similarity, optionally creates fix PRs. 5. **Agent Observability:** Built-in monitoring and telemetry for agentic systems.

Architecture Patterns: - Central CLI engine powers all surfaces (TUI, VS Code, Cloud) - HTTP + SSE communication between clients and CLI - Provider Router connects to 500+ models - Cloudflare-based cloud infrastructure - Durable Objects for state management

Tool Use: - Full MCP support - Built-in tools: file edit, shell execution, search, browser - LSP Client integration for code intelligence - Custom subagent tool

Context Management: - Memory Bank for cross-session persistence - Mode-specific context policies - Subagent isolation (separate conversation history) - Organization Modes Library for shared team modes

Edit Format: - Search/Replace blocks (XML-like) - Diff view in IDE - File timeline for tracking changes

UI Components: - VS Code extension (sidebar chat + agent manager) - JetBrains extension - Terminal TUI - Web dashboard (for cloud agents) - CLI (`kilo`, `kilo run`, `kilo serve`)

Performance Features: - Multi-session orchestration with git worktree isolation - Parallel subagent execution - Provider routing with automatic failover - Auto model tiers (Frontier/Free/Open routing)

Integration APIs: - `kilo serve` HTTP server mode - REST API + SSE streaming - `@kilocode/sdk` for custom clients - GitHub/GitLab webhooks - Telegram/Discord/Slack bots

Game Changers: 1. **5 specialized modes with subagent delegation** - Mode-specific agents that can spawn subagents create a recursive agent hierarchy 2. **Gas Town multi-agent platform** - Cloudflare-native multi-agent orchestration with formal state machine (Town→Rig→Bead→Convoy) 3. **Auto Triage + Code Review + App Builder** - Full DevOps automation pipeline built into the agent platform

Best Feature to Port: The **Subagent delegation system** - The ability for any agent to spawn isolated child agents for parallel work is a massive force multiplier. This should be a core HelixCode capability.

2.7 OpenCode (opencode-ai/opencode) - 12,329 stars

Repository Structure: - Language: Go (100%) - Size: 1.4MB (very lightweight) - Single binary, zero dependencies - Modular architecture: cmd, internal/app, internal/config, internal/db, internal/llm, internal/tui, internal/lsp - Cobra CLI framework, Bubble Tea TUI

Core Features: - 75+ model providers (most comprehensive) - Native TUI via Bubble Tea (Go) - LSP integration for code intelligence - MCP server support - Plugin system with TypeScript/JavaScript (25+ lifecycle hooks) - Multi-session support (parallel conversations) - SQLite database for persistent storage - Vim-like integrated editor - File change tracking - Named arguments for custom commands - `opencode -m` headless single-prompt execution - `opencode serve` server mode - AGENTS.md custom instructions - Variants support for different use cases

Unique Innovations: 1. **75+ Providers with Zero Lock-in:** Claude, OpenAI, Google, Groq, Fireworks, Together, OpenRouter, Azure, AWS Bedrock, Ollama, and 60+ more. Can switch mid-session. 2. **Plugin System with 25+ Hooks:** TypeScript/JavaScript plugins with hooks for session lifecycle, tool execution, context modification. Plugin ecosystem via npm. Comparable to “oh-my-zsh for AI agents” (OMO super-plugin demonstrates this). 3. **LSP + AST-grep Tools:** Real Language Server Protocol integration gives AI actual code understanding, not just text search. Diagnostics feedback loop for self-correction. 4. **Client/Server Architecture:** Run server headlessly, connect from multiple clients. Enables team sharing of sessions and context. 5. **Lightweight Go Binary:** Single binary, fast startup, minimal resource usage compared to TypeScript/Python alternatives.

Architecture Patterns: - Go modules with clean separation - Cobra for CLI, Bubble Tea for TUI - SQLite for local persistence - HTTP server for remote connections - Event-driven plugin system

Tool Use: - Built-in tools: glob, grep, ls, view, write, edit, patch, diagnostics, bash, fetch, sourcegraph, agent - MCP server integration (stdio transport) - Plugin custom tools via TypeScript

Context Management: - Session-based with SQLite persistence - Multi-session with independent context - AGENTS.md for project-specific instructions - External editor support for composing messages

Edit Format: - Search/Replace blocks - Patch application - Whole file writes - Multi-edit support

UI Components: - Bubble Tea TUI (native terminal, not web wrapper) - CLI mode for scripting - Server mode for IDE integrations - External editor integration - File browser in TUI

Performance Features: - Go binary (fast, low memory) - SQLite local storage - Streaming responses - Background task support

Integration APIs: - `opencode serve` HTTP + SSE API - Plugin API with 25+ hooks - MCP server compatibility - Custom commands with named arguments - AGENTS.md configuration

Game Changers: 1. **75+ provider support with mid-session switching** - Unmatched model flexibility eliminates vendor lock-in entirely 2. **LSP integration with diagnostics feedback loop** - Gives AI actual semantic understanding of code, not just text patterns 3. **Plugin system with 25+ lifecycle hooks** - Creates an extensible ecosystem similar to VS Code extensions but for CLI agents

Best Feature to Port: The **LSP integration with diagnostics feedback loop** - Having the agent use actual language servers for code intelligence (not just text search) is a huge leap in accuracy. The AI can get real-time type information, errors, and suggestions.

2.8 Gemini CLI (google-gemini/gemini-cli) - 103,087 stars

Repository Structure: - Language: TypeScript - Size: 93MB - Monorepo with `packages/cli/` and `packages/core/` - npm workspaces - React/Ink for TUI rendering

Core Features: - 1M token context window (Gemini 2.5 Pro) - ReAct (Reason and Act) loop architecture - Plan Mode (read-only exploration) - Multimodal input (images, PDFs, sketches) - Google Search grounding (real-time information) - Built-in developer tools: shell, file ops, web search, web fetch - GitHub integration (PR reviews, issue triage) - GitHub Actions integration - MCP support - Custom commands (slash commands) - Extensions ecosystem - `gemini.md` context files - OAuth and API key auth - Free tier: 60 req/min, 1,000 req/day

Unique Innovations: 1. **1M Token Context Window:** With Gemini 2.5 Pro, can understand entire codebases in a single context window. No need for complex indexing or retrieval systems for most projects. 2. **Plan Mode with ask_user tool:** Read-only mode where agent explores codebase, maps dependencies, proposes solutions WITHOUT modifying files. `ask_user` tool pauses research to ask targeted clarifying questions. Bidirectional communication ensures plans align with vision. 3. **Multimodal to Code:** Generate applications from hand-drawn sketches, analyze architecture diagrams, extract implementation from PDF specs. Bridges visual thinking and code. 4. **Google Search Grounding:** Real-time web search with citations for up-to-date information. Not training-cutoff limited. 5. **React/Ink TUI:** Modern terminal UI built with React component model. Component-based architecture enables rich interactive experiences in terminal.

Architecture Patterns: - Monorepo: CLI (UI) + Core (AI orchestration) - ReAct loop: Reason → Act → Observe → Repeat - React/Ink for terminal UI - Tool scheduler with confirmation gates - Modular tool registry

Tool Use: - Built-in: write-file, read-file, edit, read-many-files, grep, glob, ls, web-fetch, web-search, shell, memoryTool - MCP client for external tools - Extension tools via custom commands

Context Management: - 1M token window reduces need for complex retrieval - `gemini.md` files for project context - Conversation memory tool - Read-only MCP tools in Plan mode

Edit Format: - Built-in edit tool (search/replace) - write-file for new files - Batch file operations

UI Components: - React/Ink interactive terminal UI - Auto-completion, command history - File path references (`@path/to/file`) - Slash commands - Non-interactive mode (`-p` for scripts, `--output-format json`) - Stream-json for real-time monitoring

Performance Features: - Streaming responses - Background command execution - Efficient tool scheduling - Confirmation caching (“Always Allow”)

Integration APIs: - `--output-format json / stream-json` - Custom commands via extensions - GitHub Actions for CI/CD - MCP server connectivity

Game Changers: 1. **1M token context** - Makes complex indexing systems unnecessary for most codebases; agent can see everything at once 2. **Plan Mode with bidirectional ask_user** - The safest planning mode in any CLI agent; actively asks clarifying questions instead of making

assumptions 3. **Multimodal input (sketches→code)** - Unique capability that no other CLI agent has; enables prototyping from visual inputs

Best Feature to Port: The **Plan Mode with ask_user tool** - The bidirectional planning approach where the agent actively asks clarifying questions instead of guessing is a massive UX improvement. This prevents the “AI did the wrong thing” problem at the source.

2.9 Amazon Q Developer CLI (aws/amazon-q-developer-cli) - 1,947 stars

Repository Structure: - Language: Rust - Size: 39MB - AWS-first architecture - Fig integration (acquired by AWS) - Rust-based CLI with async runtime

Core Features: - AWS ecosystem integration (CDK, Lambda, DynamoDB, etc.) - Fig integration for command-line autocomplete - Architecture diagram generation (via aws-diagram-mcp-server) - CDK construct abstraction suggestions - Local mode with on-device processing - IDE extensions (VS Code, JetBrains) - ChatGPT plan integration (via AWS Bedrock)

Unique Innovations: 1. **Fig Integration:** Command-line autocomplete and intellisense for the terminal. Knows your CLI tools, flags, and options. This makes the terminal itself smarter before AI even gets involved. 2. **Architecture Diagram Generation:** Uses aws-diagram-mcp-server to generate AWS architecture visuals (sequence, class, flow diagrams) in Python Diagram’s format. Can “talk to your diagrams” - update CDK constructs and sync code with diagrams. 3. **CDK Construct Abstraction:** Proposes higher-level CDK constructs from low-level resource definitions, improving architecture modularity and testability. 4. **AWS-Native Context:** Deep understanding of AWS services, IAM policies, CloudFormation, CDK patterns. Not generic coding but cloud-native development.

Architecture Patterns: - Rust-based for performance - MCP server architecture for tool extensibility - AWS service integration via SDK - Fig plugin system for CLI enhancement

Tool Use: - MCP servers for extensibility - AWS-specific tools (CDK, CloudFormation, etc.) - Standard file/shell tools

Context Management: - AWS service context - Project-specific AWS configuration - CDK stack context

Edit Format: - Standard diff-based edits - CDK-specific transformations

UI Components: - Terminal CLI - IDE extensions - Fig-enhanced terminal

Performance Features: - Rust implementation - Local processing options - AWS-optimized inference routing

Integration APIs: - AWS SDK integration - MCP server protocol - Fig plugin API

Game Changers: 1. **Fig terminal intellisense** - Makes the terminal itself intelligent before AI介入; every CLI agent should have this 2. **Architecture diagram ↔ code sync** - Bidirectional relationship between diagrams and code is unique 3. **CDK construct abstraction** - AI that understands cloud architecture patterns and can improve modularity

Best Feature to Port: The **Fig-style terminal intellisense** - Enhancing the terminal with command autocomplete and context-aware suggestions BEFORE invoking AI is a fundamental UX improvement that every CLI agent should have.

2.10 GPT Engineer (AntonOsika/gpt-engineer) - 55,226 stars

Repository Structure: - Language: Python - Size: 20MB - Simple, focused architecture - `gpt_engineer/` package with CLI entry point - Prompt-based workflow

Core Features: - Full project generation from natural language prompts - Memory/prompt system for maintaining project context - Vision capabilities (image-to-code) - Multiple AI model support - Self-healing code generation - Interactive clarification loop - File-based prompt management

Unique Innovations: 1. **Prompt-Based Project Scaffolding:** Unlike other agents that modify existing code, GPT Engineer specializes in generating ENTIRE projects from scratch. The entire workflow is prompt-driven. 2. **Memory System:** File-based memory that persists project requirements, decisions, and context across sessions. Uses `.gpteng/` directory for memory storage. 3. **Interactive Clarification:** Before generating, asks clarifying questions to refine requirements. Reduces misunderstanding and rework. 4. **Self-Healing:** Attempts to fix errors automatically by re-prompting with error context.

Architecture Patterns: - Simple Python script architecture - File-based memory system - Prompt-driven workflow - Clarification → Generation → Refinement loop

Tool Use: - Standard file operations - Shell execution - No MCP (custom tools)

Context Management: - `.gpteng/memory/` for persistent context - File-based prompt storage - Project requirements documentation

Edit Format: - Whole file generation (primary use case) - Diff-based for modifications

UI Components: - Simple CLI interface - Interactive Q&A for clarification - Streaming output

Performance Features: - Lightweight Python implementation - Minimal dependencies - Fast startup

Integration APIs: - CLI interface - Python API for embedding - Custom prompt templates

Game Changers: 1. **Prompt-driven full project generation** - Specializes in greenfield development, not just editing existing code 2. **Interactive requirement clarification** - Asks questions before building, dramatically improving output quality 3. **File-based memory system** - Simple but effective persistent context without complex databases

Best Feature to Port: The **Interactive Clarification Loop** - Asking clarifying questions BEFORE generating code is a simple but extremely effective pattern that dramatically reduces rework. Every agent should do this.

3. Feature Comparison Matrix

3.1 Repository & Language

Agent	Stars	Language	License	Size	Age	Contributors
Gemini CLI	103,087	TypeScript	Apache-2.0	93MB	~1 year	Google team
	79,852	Rust (95.6%)	Apache-2.0	423MB		400+

Agent	Stars	Language	License	Size	Age	Contributors
OpenAI Codex					1 year	
Cline	61,341	TypeScript	Apache-2.0	390MB	2 years	Large community
GPT Engineer	55,226	Python	MIT	20MB	2+ years	Community
Aider	44,304	Python	Apache-2.0	140MB	2+ years	Active
Kilo Code	18,881	TypeScript	MIT	407MB	1 year	GitLab founder
Plandex	15,326	Go	MIT	57MB	2 years	Solo founder
OpenCode	12,329	Go	MIT	1.4MB	1 year	SST team
Forge	7,177	Rust	Apache-2.0	18MB	1 year	Community
Amazon Q	1,947	Rust	Apache-2.0	39MB	2 years	AWS team

3.2 Core Capabilities

Capability	Aider	Cline	Codex	Plandex	Forge	Kilo	OpenCode	Gemini	Amazon Q
File Read	✓	✓	✓	✓	✓	✓	✓	✓	✓
File Write	✓	✓	✓	✓	✓	✓	✓	✓	✓
Shell Execute	✓	✓	✓	✓	✓	✓	✓	✓	✓
Git Integration	✓(native)	✓	✓	✓	✓	✓	✓	✓	✓
Browser Use	✓(Play)	✓(Puppet)	✗	✗	✗	✓	✗	✗	✗
Code Search	✓	✓	✓	✓	✓	✓	✓	✓	✓
LSP Integration	✓	✗	✗	✗	✗	✓	✓	✗	✗
MCP Support	✗	✓	✓	✗	✓	✓	✓	✓	✓
Plan Mode	✓(Arch)	✓	✗	✓	✓	✓	✗	✓	✗
Voice Input	✓	✗	✗	✗	✗	✗	✗	✗	✗
Vision/ Multimodal	✓	✗	✗	✗	✗	✗	✗	✓	✗

3.3 Model Support

Provider	Aider	Cline	Codex	Plandex	Forge	Kilo	OpenCode	Gemini	Amazon Q
OpenAI/ GPT	✓	✓	✓(native)	✓	✓	✓	✓	✗	✗
Anthropic/ Claude	✓	✓	✗	✓	✓	✓	✓	✗	✓
Google/ Gemini	✓	✓	✗	✗	✓	✓	✓	✓(native)	✗
Local/ Ollama	✓	✓	✓(—oss)	✗	✓	✓	✓	✗	✗
DeepSeek	✓	✓	✗	✗	✓	✓	✓	✗	✗
OpenRouter	✓	✓	✗	✓	✗	✓	✓	✗	✗
AWS Bedrock	✗	✓	✗	✗	✗	✓	✓	✗	✓
BYOK Support	✓	✓	✓	✓	✓	✓	✓	✓	✗
# Providers	20+	30+	1+(—oss)	5+	5+	500+	75+	1	2

3.4 Context Management

Approach	Aider	Cline	Codex	Plandex	Forge	Kilo	OpenCode	Gemini	Amazon Q
Context Window	200K	200K	200K	2M	200K	200K	200K	1M	200K
Repo Map	✓(TS)	✓	✗	✓(TS)	✓	✓	✗	✗	✗
Embeddings/ Vector	✗	✓	✗	✗	✓	✓	✗	✗	✗
Tree-sitter Parsing	✓	✓	✗	✓	✓	✓	✓	✗	✗
Auto-compaction	✗	✗	✓	✗	✓	✓	✗	✗	✗
Cross-session Memory	✗(git)	✓(MB)	✗	✗	✓	✓(MB)	✓(DB)	✗	✗
Incremental Indexing	✗	✓	✗	✓	✓	✓	✗	✗	✗

3.5 Edit Format

Format	Aider	Cline	Codex	Plandex	Forge	Kilo	OpenCode	Gemini	Amazon Q	En
Search/Replace	✓	✓	✗	✓	✓	✓	✓	✓	✓	
Unified Diff	✓	✗	✓	✓	✓	✗	✗	✗	✗	
Whole File	✓	✗	✗	✗	✓	✗	✓	✗	✗	✓(
Patch Apply	✗	✗	✓	✗	✓	✗	✓	✗	✗	
Fuzzy Matching	✓(4-layer)	✗	✗	✗	✗	✗	✗	✗	✗	
Sandboxed Review	✗	✓(CP)	✗	✓	✗	✗	✗	✗	✗	

CP = Checkpoints (Shadow Git)

3.6 UI & Interaction

Feature	Aider	Cline	Codex	Plandex	Forge	Kilo	OpenCode	Gemini	Amazon Q
Terminal TUI	✓	✓(CLI2)	✓	✓	✓	✓	✓(BT)	✓(Ink)	✓
IDE Extension	✓	✓(VS)	✓	✗	✗	✓	✓	✗	✓
Web Interface	✗	✗	✗	✗	✓(dash)	✓	✗	✗	✗
Plan Mode UI	✓	✓	✗	✓	✓	✓	✗	✓	✗
Streaming Output	✓	✓	✓	✓	✓	✓	✓	✓	✓
Headless/CI Mode	✓	✓(—y)	✓	✗	✗	✓	✓	✓	✗
Multi-session	✗	✗	✗	✗	✓	✓	✓	✗	✗

BT = Bubble Tea, Ink = React/Ink, VS = VS Code, dash = dashboard

3.7 Security & Sandboxing

Feature	Aider	Cline	Codex	Plandex	Forge	Kilo	OpenCode	Gemini	Amazon Q
OS-level Sandbox	✗	✗	✓(Seatbelt/Docker)	✗	✗	✗	✗	✗	✗

Feature	Aider	Cline	Codex	Plandex	Forge	Kilo	OpenCode	Gemini	Amazon Q
Approval Gates	✓	✓(per-step)	✓(tiered)	✓	✓	✓	✓	✓	✓
Command Filtering	✗	✗	✓	✗	✗	✗	✗	✗	✗
Network Restrictions	✗	✗	✓(iptables)	✗	✗	✗	✗	✗	✗
Shadow Git	✗	✓	✗	✗	✗	✗	✗	✗	✗
Cumulative Diff	✗	✗	✗	✓	✗	✗	✗	✗	✗
ZDR Compliance	✗	✗	✓	✗	✗	✗	✗	✗	✗

3.8 Multi-Agent & Orchestration

Feature	Aider	Cline	Codex	Plandex	Forge	Kilo	OpenCode	Gemini	Amazon Q	Er
Subagents	✗	✗	✗	✗	✓	✓	✓(OMO)	✗	✗	
Multi-agent Patterns	✗	✗	✗	✗	✓(6)	✓(Gas)	✗	✗	✗	
Agent Hierarchy	✗	✗	✗	✗	✓	✓	✗	✗	✗	
Quality Scoring	✗	✗	✗	✗	✓	✗	✗	✗	✗	
A/B Testing	✗	✗	✗	✗	✓	✗	✗	✗	✗	
Auto Triage	✗	✗	✗	✗	✗	✓	✗	✗	✗	
Auto Code Review	✗	✗	✗	✗	✗	✓	✗	✗	✗	

3.9 Performance & Optimizations

Feature	Aider	Cline	Codex	Plandex	Forge	Kilo	OpenCode	Gemini	Amazon Q
Prompt Caching	✓	✗	✓	✗	✗	✗	✗	✗	✗
Rust Implementation	✗	✗	✓	✗	✓	✗	✗	✗	✓
Go Implementation	✗	✗	✗	✓	✗	✗	✓	✗	✗

Feature	Aider	Cline	Codex	Plandex	Forge	Kilo	OpenCode	Gemini	Amazon Q
Streaming	✓	✓	✓	✓	✓	✓	✓	✓	✓
Background Tasks	✗	✓	✗	✗	✗	✓	✓	✓	✗
Model Switching	✓	✓ ^(P/A)	✗	✗	✓	✓	✓	✗	✗
Auto-compaction	✗	✗	✓	✗	✓	✗	✗	✗	✗

4. Architecture Pattern Analysis

4.1 The Agent Loop Pattern

All 10 agents share a common execution structure (the “Agent Loop”):

```
User Input → Construct Prompt → LLM Inference →
├─ Tool Call → Execute Tool → Observe Result → [Loop]
└─ Assistant Message → Present to User → [End Turn]
```

Variations: - **Aider**: Tight git integration loop; every edit committed - **Cline**: Plan/Act state machine within loop; checkpoints between iterations - **Codex**: Stateless requests with careful prefix preservation; ZDR compliance - **Forge**: Multi-agent orchestrator loop with 6 patterns - **Kilo**: Mode-specific loops with subagent delegation

4.2 Context Management Architectures

Pattern	Agents	Description
Manual Control	Aider	User explicitly adds/drops files via / add, /drop
Full-Text Indexing	Plandex, OpenCode	Tree-sitter parsing + indexing for large codebases
Vector Search	Cline, Forge, Kilo	Embeddings + semantic search for relevant context
Massive Context	Gemini, Plandex	1M-2M token windows reduce need for retrieval
Memory Bank	Cline, Kilo	Structured files for cross-session persistence
Git-as-Memory	Aider	Commit history as persistent context
Auto-compaction	Codex	Automatic summarization with encrypted latent understanding

4.3 Edit Format Evolution

The industry has converged on several edit formats, each with tradeoffs:

Format	Token Efficiency	Model Compatibility	Robustness	Best For
Search/Replace	High	Most models	Medium	General purpose
Unified Diff	High	GPT-4 Turbo, Codex	High	Complex changes
Whole File	Low	All models	High	Small files, weak models
Patch	High	Codex	Medium	Batch changes

Aider's Innovation: 4-layer fuzzy matching (exact → whitespace → indentation → diff) makes search/replace robust even with imperfect AI output.

4.4 Security Model Spectrum

Level	Implementation	Agents
None	Trust the user	GPT Engineer, early agents
Approval Gates	Ask before execute	Cline, Aider, OpenCode, Gemini
Tiered Approval	Auto-approve reads, ask for writes	Codex (Suggest/Auto-Edit/Full Auto)
Shadow Git	Independent rollback checkpoint	Cline
Cumulative Diff	Sandbox all changes before apply	Plandex
OS Sandbox	Seatbelt/Docker/iptables	Codex

5. Game-Changing Features

5.1 Top 3 Game-Changers Per Agent

Aider

1. **Architect/Editor Dual-Model** - Separating reasoning from editing (85% pass rate with o1-preview + o1-mini)
2. **4-Layer Fuzzy Matching** - Exact → Whitespace → Indentation → diff for robust edits
3. **Git-Native Workflow** - Every change auto-committed, full audit trail

Cline

1. **Plan/Act with Different Models** - Use cheap models for execution, expensive for planning
2. **Shadow Git Checkpoints** - Granular rollback preserving full environment state
3. **Computer Use / Browser Automation** - Real browser verification of UI changes

OpenAI Codex

1. **OS-Native Sandboxed Execution** - Seatbelt (macOS) / Docker+iptables (Linux) / Native (Windows)
2. **Automatic Context Compaction** - /responses/compact with encrypted latent understanding preservation
3. **Stateless ZDR Architecture** - Zero Data Retention compliance without losing functionality

Plandex

1. **Cumulative Diff Review Sandbox** - Accumulate ALL changes, review BEFORE applying ANY
2. **2M Token Context + 20M Token Indexing** - Massive context handling for large projects
3. **Version-Controlled Model Packs** - Model config as versioned artifacts

Forge

1. **6 Orchestration Patterns** - Sequential, Parallel, Leader-Worker, Dynamic Routing, Explorer-Critic, Kanban
2. **Quality Scoring with Automated Gates** - 100pt scale for CI/CD integration
3. **A/B Testing for Agent Configs** - Data-driven optimization of prompts and models

Kilo Code

1. **5 Specialized Modes + Subagent Delegation** - Recursive agent hierarchy with isolated subagents
2. **Gas Town Multi-Agent Platform** - Cloudflare-native with formal state machine (Town→Rig→Bead→Convoy)
3. **Auto Triage + Auto Review + App Builder** - Full DevOps automation pipeline

OpenCode

1. **75+ Provider Support with Mid-Session Switching** - Unmatched model flexibility
2. **LSP Integration with Diagnostics Feedback** - Semantic code understanding, not text search
3. **Plugin System with 25+ Lifecycle Hooks** - Extensible ecosystem via TypeScript/npm

Gemini CLI

1. **1M Token Context Window** - Eliminates need for complex indexing for most projects
2. **Plan Mode with ask_user Tool** - Bidirectional planning with active clarification
3. **Multimodal Input (Sketches→Code)** - Visual prototyping to functional code

Amazon Q

1. **Fig-Style Terminal Intellisense** - Context-aware autocomplete before AI介入
2. **Architecture Diagram  Code Sync** - Bidirectional diagrams and code relationship
3. **CDK Construct Abstraction** - AI that understands cloud architecture patterns

GPT Engineer

1. **Interactive Clarification Loop** - Asks questions before building
 2. **Prompt-Driven Full Project Generation** - Greenfield development from natural language
 3. **File-Based Memory System** - Simple persistent context without databases
-

6. Recommendations for HelixCode

6.1 ABSOLUTE BEST Features to Port (Priority Order)

Tier 1: Must Have

1. **Plan/Act Dual-Mode with Model Switching** (from Cline)
 - Read-only Plan mode for exploration and architecture
 - Act mode for implementation
 - Different models for each mode (cost/quality optimization)
 - Context carries over between modes
 - **Impact:** Dramatically improves control, safety, and cost-efficiency
2. **OS-Native Sandboxed Execution** (from Codex)
 - macOS Seatbelt profiles
 - Linux Docker containers with iptables
 - Windows native sandbox
 - File scope limiting + network restrictions
 - **Impact:** Essential for trustworthy autonomous agents; prevents catastrophic mistakes
3. **Architect/Editor Dual-Model Architecture** (from Aider)
 - Architect model reasons about solution
 - Editor model converts to precise file edits
 - 85% pass rate demonstrated
 - **Impact:** Separates reasoning from precision; any model can be paired with a strong editor

Tier 2: Strongly Recommended

1. **Cumulative Diff Review Sandbox** (from Plandex)
 - All AI changes accumulate in sandbox
 - Review ALL before applying ANY
 - Atomic apply or reject
 - **Impact:** Prevents “AI rewrote half my project” disasters
2. **LSP Integration with Diagnostics Feedback Loop** (from OpenCode)
 - Real language server protocol integration
 - Type information, real-time errors
 - AI self-correction via diagnostics
 - **Impact:** Massive improvement in code accuracy vs text search
3. **Subagent Delegation System** (from Kilo Code)
 - Any agent can spawn isolated child agents
 - Parallel execution of subtasks
 - Automatic task decomposition
 - **Impact:** Force multiplier; enables complex workflows impossible for single agents
4. **Interactive Clarification Loop** (from Gemini CLI + GPT Engineer)
 - Agent asks clarifying questions before acting
 - Bidirectional communication during planning

- Reduces misunderstanding and rework
- **Impact:** Dramatically reduces “AI did the wrong thing” problems

Tier 3: Highly Valuable

- 1. **6 Orchestration Patterns** (from Forge)
 - Sequential, Parallel, Leader-Worker, Dynamic Routing, Explorer-Critic, Kanban
 - Enables complex multi-agent workflows
 - **Impact:** Future-proofs HelixCode for advanced agent coordination
- 2. **Shadow Git Checkpoints** (from Cline)
 - Independent Git repository for AI state
 - Granular rollback without polluting main history
 - Preserves terminal and browser state
 - **Impact:** Safety net for experimentation
- 3. **75+ Provider Router** (from OpenCode)
 - BYOK with any provider
 - Mid-session model switching
 - Automatic failover
 - **Impact:** Eliminates vendor lock-in
- 4. **Tree-Sitter + Vector Indexing** (from Plandex/Aider)
 - AST-based code understanding
 - Semantic search via embeddings
 - Incremental updates
 - **Impact:** Accurate codebase comprehension at scale
- 5. **Fig-Style Terminal Intellisense** (from Amazon Q)
 - Command autocomplete before AI介入
 - Context-aware suggestions
 - **Impact:** Improves baseline terminal UX

6.2 Recommended HelixCode Architecture

Based on this analysis, the optimal HelixCode architecture should combine:

HELIXCODE ARCHITECTURE
UI Layer: Terminal TUI (Bubble Tea) + IDE Extension + Web Dashboard
Mode Layer: Plan Mode (read-only) ↔ Act Mode (execute) + Architect/Editor dual-model option
Agent Layer: Primary Agent + Subagent Delegation + Multi-agent orchestration patterns
Tool Layer: Built-in tools + MCP servers + LSP integration + Custom plugins (25+ hooks)
Context Layer: Tree-sitter indexing + Vector search + 2M token window + Auto-compaction + Memory Bank (cross-session) + Git-as-memory
Edit Layer: Search/Replace + Unified Diff + Patch + 4-layer fuzzy matching + Cumulative sandbox

Security Layer: OS sandbox (Seatbelt/Docker) + Tiered approval + Shadow Git checkpoints + Command filtering + Network restrictions
Provider Layer: 75+ provider router with BYOK + Mid-session model switching + Auto failover
Platform: Go (performance) or Rust (safety) core + SQLite persistence + HTTP/SSE server API

6.3 Implementation Priority

Phase 1 (Core): 1. Plan/Act dual-mode 2. OS-native sandboxing 3. Architect/Editor dual-model 4. LSP integration 5. Tree-sitter indexing

Phase 2 (Advanced): 6. Subagent delegation 7. Cumulative diff sandbox 8. Interactive clarification 9. Shadow Git checkpoints 10. 75+ provider router

Phase 3 (Ecosystem): 11. 6 orchestration patterns 12. Plugin system (25+ hooks) 13. Multi-agent platform (Gas Town-style) 14. Auto Triage / Review / Builder 15. Fig-style terminal intellisense

7. Summary Statistics

Most Common Patterns Across All Agents

- **Git integration:** 10/10 agents
- **Shell execution:** 10/10 agents
- **Streaming output:** 10/10 agents
- **Approval gates:** 10/10 agents
- **MCP support:** 5/10 agents (growing)
- **Plan mode:** 5/10 agents
- **Multi-session:** 3/10 agents
- **OS sandbox:** 1/10 agents (Codex only - huge opportunity)
- **Subagent delegation:** 2/10 agents
- **LSP integration:** 2/10 agents

Technology Stack Distribution

- **Python:** Aider, GPT Engineer (2)
- **TypeScript:** Cline, Gemini CLI, Kilo Code (3)
- **Rust:** Codex, Forge, Amazon Q (3)
- **Go:** Plandex, OpenCode (2)

License Distribution

- **Apache-2.0:** Aider, Cline, Codex, Forge, Gemini, Amazon Q (6)
 - **MIT:** Plandex, Kilo Code, OpenCode, GPT Engineer (4)
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Report generated from analysis of 10 CLI AI coding agent repositories, documentation, and architecture sources. All star counts, features, and capabilities accurate as of May 2026.